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Self-Supervised Learning for 3D Non-Cartesian Multi-Coil MRI  
Reconstruction: an SSDU Pipeline for the NC-PDNet Unrolled Network

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## Abstract

Magnetic resonance imaging is limited by the time it takes to acquire data, and high spatial resolution makes that time prohibitive. Acquisitions are therefore undersampled, and the missing information is recovered by reconstruction algorithms. Deep learning has become the dominant approach to this problem, but in its usual supervised form it requires fully sampled reference images, which cannot be obtained at high resolution in three dimensions: acquiring them would take longer than a patient can tolerate. Self-supervised methods remove this requirement by holding back part of the acquired measurements and asking the network to predict them. SSDU, Self-Supervised learning via Data Undersampling, implements this by splitting the acquired k-space into two disjoint subsets, one given to the network and one used to compute the loss. The method was designed for Cartesian acquisitions, where the measurements lie on a regular grid. This internship, carried out at CEA NeuroSpin within the Inria-CEA MIND team, transposed it to non-Cartesian three-dimensional multi-coil acquisitions using optimised SPARKLING trajectories, with the NC-PDNet unrolled network. Since the samples follow continuous curves rather than a grid, the split had to be redefined on the trajectory itself. A stratified procedure was developed that reproduces the sampling density of the full trajectory without assuming any particular shape for it, an improvement of two orders of magnitude over a per-shot selection. The pipeline trains end to end and produces anatomically plausible volumes without any reference image. Trained on a single acquisition, it overfits, and the effect of the split ratio proved indistinguishable from run-to-run variability. Training on multiple patients is the next step.

**Keywords:** magnetic resonance imaging, reconstruction, self-supervised learning, non-Cartesian sampling, deep learning

## Résumé

L'imagerie par résonance magnétique est limitée par son temps d'acquisition, que la résolution spatiale rend rapidement prohibitif. Les acquisitions sont donc sous-échantillonnées, et l'information manquante est restituée par des algorithmes de reconstruction. L'apprentissage profond s'est imposé pour cette tâche, mais sa forme supervisée usuelle exige des images de référence entièrement échantillonnées, impossibles à obtenir en haute résolution tridimensionnelle : leur acquisition durerait plus longtemps qu'un patient ne peut rester immobile. Les méthodes auto-supervisées lèvent cette contrainte en réservant une partie des mesures acquises et en demandant au réseau de les prédire. SSDU met en œuvre cette idée en partitionnant le k-space acquis en deux sous-ensembles disjoints, l'un fourni au réseau, l'autre servant à calculer la fonction de coût. La méthode a été conçue pour des acquisitions cartésiennes, où les mesures occupent une grille régulière. Ce stage, réalisé au CEA NeuroSpin au sein de l'équipe Inria-CEA MIND, l'a transposée à des acquisitions non-cartésiennes tridimensionnelles multi-antennes suivant des trajectoires SPARKLING optimisées, avec le réseau déroulé NC-PDNet. Les échantillons suivant des courbes continues et non une grille, le partitionnement a dû être redéfini sur la trajectoire elle-même. Une procédure stratifiée a été développée, qui reproduit la densité d'échantillonnage de la trajectoire complète sans présupposer sa forme, avec un gain de deux ordres de grandeur sur une sélection par shot. Le pipeline s'entraîne de bout en bout et produit des volumes anatomiquement plausibles sans aucune image de référence. Entraîné sur une seule acquisition, il surapprend, et l'effet du ratio de partition s'est révélé indiscernable de la variabilité entre exécutions. L'entraînement multi-patients constitue l'étape suivante.

**Mots-clés :** imagerie par résonance magnétique, reconstruction, apprentissage auto-supervisé, échantillonnage non-cartésien, apprentissage profond

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# List of Abbreviations and Variables

## Abbreviations

| Abbreviation | Definition                                                         |
|--------------|--------------------------------------------------------------------|
| AP-HP        | Assistance Publique – Hôpitaux de Paris                            |
| CCRT         | Centre de Calcul Recherche et Technologie                          |
| CEA          | Commissariat à l'énergie atomique et aux énergies alternatives     |
| COR          | Reference reconstruction computed with off-resonance correction    |
| DDSS         | Dual-Domain Self-Supervised learning                               |
| FOV          | Field of View                                                      |
| GPU          | Graphics Processing Unit                                           |
| ILIS         | Institut Lillois d'Ingénierie de la Santé                          |
| MIND         | Models and Inference for Neuroimaging Data                         |
| MRI          | Magnetic Resonance Imaging                                         |
| NC-PDNet     | Non-Cartesian Primal-Dual Network                                  |
| NOC          | Reference reconstruction computed without off-resonance correction |
| NUFFT        | Non-Uniform Fast Fourier Transform                                 |
| PSNR         | Peak Signal-to-Noise Ratio                                         |
| RF           | Radio-Frequency                                                    |
| SPARKLING    | Spreading Projection Algorithm for Rapid K-space sampling          |
| SSDU         | Self-Supervised learning via Data Undersampling                    |
| SSIM         | Structural Similarity Index Measure                                |
| SWI          | Susceptibility Weighted Imaging                                    |
| ZS-SSL       | Zero-Shot Self-Supervised Learning                                 |

## Terms

| Term                 | Definition                                                                                                                       |
|----------------------|----------------------------------------------------------------------------------------------------------------------------------|
| Aliasing             | Artefact caused by sampling below the Nyquist rate, where parts of the object outside the field of view fold back into the image |
| Coil                 | Receiver antenna placed around the head                                                                                          |
| Density compensation | Weighting of each sample by the inverse of the local sampling density, applied before the adjoint transform                      |
| Gradient             | Magnetic field varying linearly in space, used to encode position into frequency                                                 |
| Ill-posed            | Property of an inverse problem admitting infinitely many solutions or amplifying noise                                           |
| Incoherence          | Property of an undersampling pattern whose artefacts are noise-like rather than structured                                       |
| k-space              | Spatial frequency domain in which the scanner acquires its measurements                                                          |
| Prior                | Assumption about what a plausible image looks like, supplied to compensate for missing data                                      |
| Sensitivity map      | Spatial weighting through which a given coil sees the object                                                                     |
| Shot                 | One excitation of the magnetisation followed by one continuous readout                                                           |
| Slew rate            | Rate at which the amplitude of a gradient can change                                                                             |
| Sparse               | Admitting a representation with few non-zero coefficients in a suitable transform domain                                         |
| Spin                 | Hydrogen nucleus whose collective magnetisation produces the measured signal                                                     |
| Trajectory           | Path followed through k-space during acquisition                                                                                 |
| Unrolled network     | Network whose layers reproduce a fixed number of iterations of an optimisation algorithm                                         |
| Voxel                | Three-dimensional counterpart of a pixel                                                                                         |

## Mathematical Symbols and Variables

| Symbol                | Definition                                                   |
|-----------------------|--------------------------------------------------------------|
| $\mathbf{x}$          | Image to be reconstructed, $\mathbf{x} \in \mathbb{C}^N$     |
| $\mathbf{y}$          | Acquired k-space measurements, $\mathbf{y} \in \mathbb{C}^M$ |
| $\mathbf{y}_\ell$     | Measurements acquired by coil $\ell$                         |
| $\mathbf{r}$          | Spatial position in the image domain                         |
| $\mathbf{k}$          | Spatial frequency, i.e. a position in k-space                |
| $\Delta k$            | Spacing between two neighbouring k-space samples             |
| $N$                   | Number of voxels in the image                                |
| $M$                   | Number of acquired k-space samples                           |
| $L$                   | Number of receiver coils                                     |
| $\Omega$              | Full set of acquired k-space locations                       |
| $\Theta$              | Subset of $\Omega$ given to the network as input             |
| $\Lambda$             | Subset of $\Omega$ held out to compute the loss              |
| $\mathbf{S}_\ell$     | Sensitivity map of coil $\ell$                               |
| $\mathcal{F}_\Omega$  | Fourier operator evaluated at the locations $\Omega$         |
| $\mathcal{A}_\Lambda$ | Forward operator evaluated at the locations $\Lambda$        |
| $\mathcal{R}$         | Regulariser                                                  |
| $f_\theta$            | Reconstruction network with weights $\theta$                 |
| $\mathcal{L}$         | Training loss                                                |
| $\Delta B_0$          | Off-resonance field map, in Hz                               |
| $TE$                  | Echo time                                                    |
| $TR$                  | Repetition time                                              |
| $T_{obs}$             | Readout duration                                             |

# 1 Introduction

Magnetic resonance imaging (MRI) is slow. A scanner does not photograph the body: it measures the object in the spatial frequency domain, sample by sample, and every sample costs time. Resolution makes this expensive, since halving the size of a voxel in all three directions multiplies the number of samples required by eight. A patient can lie still for a few minutes at best, which sets a hard limit on what can be acquired, and beyond which motion degrades the image as surely as missing data would.

The response has been to acquire less and to compute more. Undersampled acquisitions, combined with reconstruction algorithms that compensate for what was not measured, have made high-resolution imaging feasible in clinically acceptable times, and deep learning has become the dominant tool for that compensation. It comes with a requirement that is rarely met in practice: training a reconstruction network in the usual way needs pairs of undersampled acquisitions and the fully sampled images they should have produced. At high resolution in three dimensions, such references would themselves take longer to acquire than any patient can tolerate. Asking for them in order to learn how to avoid acquiring them is circular, and it is what currently keeps learned reconstruction away from clinical protocols.

Self-supervised methods remove that requirement. Rather than comparing a reconstruction to a reference image, they hold back part of the acquired measurements and ask the network to predict them. SSDU, Self-Supervised learning via Data Undersampling, formalises this by splitting the acquired k-space into two disjoint subsets, one given to the network and one used to compute the loss. It was designed and validated for Cartesian acquisitions, where the measurements lie on a regular grid.

The acquisitions studied in this work do not. They use SPARKLING, a family of optimised non-Cartesian trajectories developed at NeuroSpin, in which the samples follow continuous curves grouped into shots. There is no grid to partition, and how the split should be defined in this setting is left open by the original work. My assignment was to build a complete SSDU pipeline for 3D multi-coil non-Cartesian reconstruction with the NC-PDNet unrolled network, to investigate how the splitting strategy should be adapted, and to evaluate the result against the reconstructions available for the dataset.

The report is organised as follows. Section 2 presents the host institution and the context of the internship. Section 3 introduces MRI acquisition, the reconstruction problem, and the self-supervised approach the work builds on. Section 4 describes the dataset, the pipeline and the protocols used. Section 5 reports the results, and Section 6 discusses what they establish, what limits them, and what remains to be done.

## 2 Host Institution and Context of the Internship

### 2.1 CEA NeuroSpin

I carried out my internship at NeuroSpin, a research centre of the French Alternative Energies and Atomic Energy Commission (CEA) located on the Paris-Saclay campus in Gif-sur-Yvette. NeuroSpin belongs to the Joliot Institute, which gathers the CEA laboratories working on life sciences, and it is the largest ultra-high field MRI facility in France, dedicated to cognitive and clinical neuroscience.

What makes the centre unusual is the scale of its scanners. NeuroSpin operates human MRI systems at 3T, 7T and 11.7T. Higher magnetic fields give access to finer spatial resolution, but they also make image acquisition longer and more sensitive to artefacts. This tension between resolution and scan time is precisely the problem my internship addresses.

### 2.2 The Inria-CEA MIND team

Within NeuroSpin, I joined the MIND team (Models and Inference for Neuroimaging Data), a joint Inria-CEA project-team created in April 2022 as a spin-off of the former Parietal team. MIND brings together permanent researchers, post-doctoral fellows, PhD students and engineers, and is led by Philippe Ciuciu, Research Director at CEA.

The team works at the intersection of statistics, machine learning and signal processing, with the brain as its application domain. Among its core methodological interests is the resolution of inverse problems in three and four dimensions, linear or non-linear, informed by biophysical models. MRI reconstruction belongs to this family of problems, and it is the one I worked on.

An MRI examination proceeds in two stages. The scanner does not measure an image directly: it samples the spatial frequency domain of the object, known as k-space, point by point along trajectories imposed by the magnetic field gradients. This is the acquisition stage, and it is what makes MRI slow, since every sample costs time. An image is then computed from these measurements, which is the reconstruction stage. Scan time can therefore be reduced from either end, by sampling k-space more efficiently or by recovering a good image from fewer samples, and MIND works on both.

On the acquisition side, the team proposed SPARKLING (Spreading Projection Algorithm for Rapid K-space sampLING), a method for generating non-Cartesian sampling trajectories, first for 2D imaging and later extended to full 3D [1]. On the reconstruction side, it benchmarked NC-PDNet, an unrolled network designed for 3D multi-coil non-Cartesian data [2]. My internship sits at the junction of these two lines of work.

I was supervised by Philippe Ciuciu and by Chaithya G. R., a research engineer in the team who works on non-Cartesian sampling and on the `mri-nufft` library.

## 3 Scientific and Technical Background

### 3.1 MRI acquisition and k-space

An MRI scanner does not take a picture. A camera collects light coming from each point of a scene separately, so the image exists as soon as the shutter closes. An MRI scanner measures something else entirely: at every instant it records a single number, and that number depends on the entire object at once. The image does not exist in the machine. It has to be computed from these measurements, and the whole difficulty of this report lies in that computation.

The signal comes from the hydrogen nuclei of the body, mostly those of water. Placed in a strong magnetic field, they behave like tiny compass needles and line up with it. These are the *spins*. A brief radio pulse tips them out of alignment, and as they return to it they emit a faint radio signal, picked up by antennas placed around the head and called *coils*. On its own this signal says nothing about where in the brain it came from: every spin contributes to it at the same time.

Localisation is obtained by making the spins sing at different pitches depending on where they are. Additional magnetic fields, called *gradients*, are superimposed on the main one. Their strength varies linearly across space, so a spin at the front of the head experiences a slightly different field from a spin at the back, and therefore emits at a slightly different frequency. The measured signal becomes a sum of frequencies, and each frequency corresponds to a position. Separating them is a job for the Fourier transform.

The signal recorded during a readout is therefore not an image but a collection of frequencies. Written for a continuous object, it takes the form

$$y(\mathbf{k}) = \int_{\mathbb{R}^3} x(\mathbf{r}) e^{-2i\pi\mathbf{k}\cdot\mathbf{r}} d\mathbf{r}, \quad (1)$$

where  $x(\mathbf{r})$  is the image to be recovered at position  $\mathbf{r}$ , and  $\mathbf{k}$  is a spatial frequency. This is a Fourier transform. The scanner measures the object in the frequency domain, which MRI calls *k-space*, and the image is obtained from it by an inverse transform. Figure 1 shows the two faces of the same brain.

The samples are not collected all at once. Where the acquisition sits in k-space at a given instant is set by the time integral of the gradients, so playing out a gradient waveform amounts to travelling along a path through k-space. That path is called a *trajectory*, and the sequence designer chooses it. Figure 2 shows the correspondence for the Cartesian case: one excitation, one gradient waveform, one line of k-space. The whole volume is covered by repeating this pattern line by line and plane by plane, which is why the number of repetitions, and therefore the scan time, grows so quickly with resolution. Nothing in the physics, however, requires the path to be a straight line, and Section 3.2 exploits that freedom.

In practice nothing is continuous. The signal is sampled at a finite number of positions, and the image lives on a discrete grid. From now on  $\mathbf{x} \in \mathbb{C}^N$  denotes the image to be reconstructed,

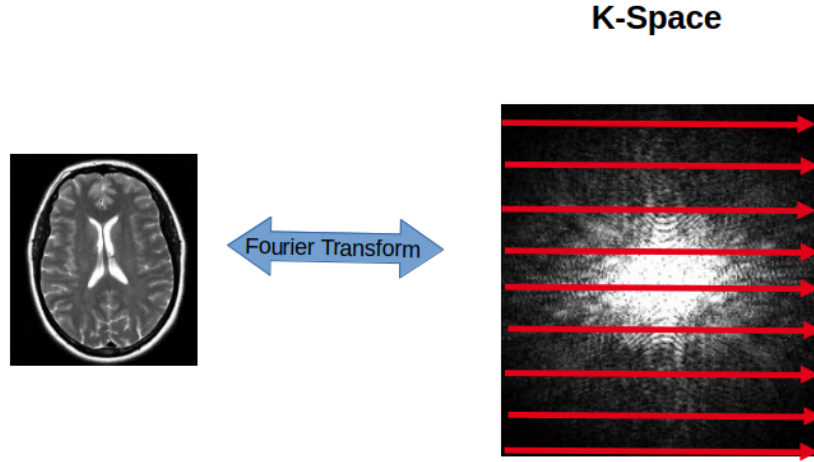


Figure 1: An MR image and its k-space: two representations of the same object, related by a Fourier transform. Most of the signal energy lies at the centre of k-space, which appears bright, while the periphery is dark. The red arrows show the order in which a Cartesian acquisition fills k-space, one horizontal line per excitation. This is the simplest sampling scheme; Section 3.2 introduces the non-Cartesian trajectories used in this work. Figure courtesy of Chaithya G. R.

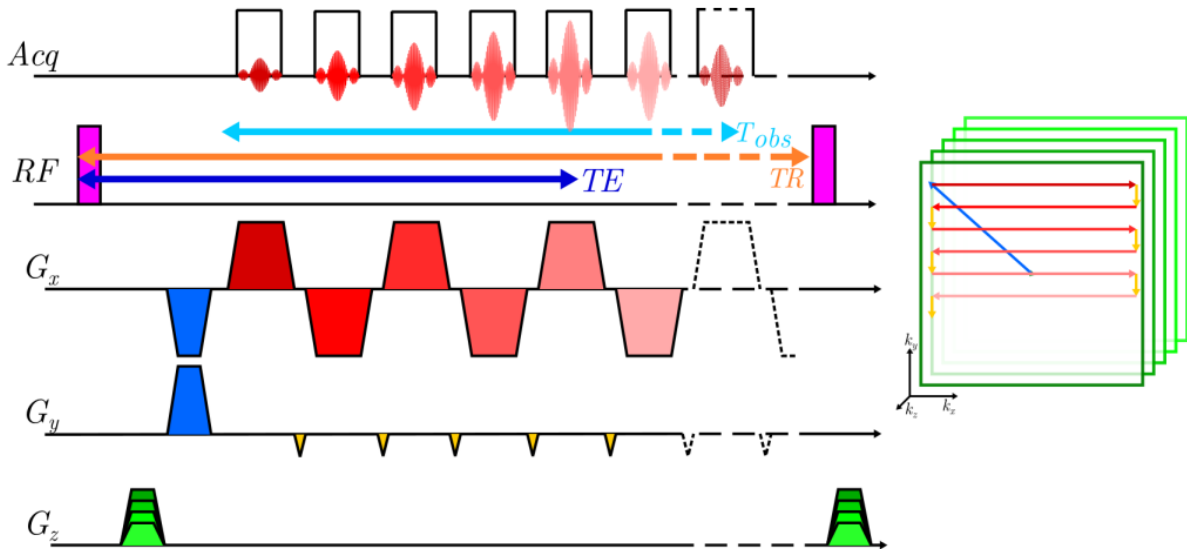


Figure 2: From gradient waveforms to a k-space trajectory, for a Cartesian acquisition. Left: a radio-frequency pulse (RF) tips the spins, then the gradients  $G_x$ ,  $G_y$  and  $G_z$  are played out while the signal is recorded during the acquisition windows (Acq).  $TE$  is the echo time,  $TR$  the repetition time between two excitations, and  $T_{obs}$  the readout duration. Right: the resulting path through k-space, one line per repetition, the volume being covered plane by plane. Figure courtesy of Chaithya G. R.

made of  $N$  *voxels*, the three-dimensional counterpart of pixels, and  $\mathbf{y} \in \mathbb{C}^M$  the  $M$  samples actually acquired. Equation (1) then becomes a linear system relating the two. Both are complex-valued, and this is not a mathematical convenience: the phase of the MR signal carries physical information, which is exactly what susceptibility weighted imaging exploits. It is also one reason why networks designed for ordinary photographs cannot be reused as they are.

Everything so far has ignored the constraint that makes MRI difficult: time. A readout lasts a few tens of milliseconds, and the sequence must then wait for the magnetisation to recover before the next excitation. Scan time therefore scales with the number of repetitions, and that number is set by the resolution. Halving the voxel size in all three directions multiplies the number of samples required by eight. A patient, meanwhile, can lie still for a few minutes at best, and every extra minute increases the risk of motion blurring the image. This is the tension the rest of this chapter addresses: how to acquire fewer samples and still recover a usable image.

### 3.2 Accelerating acquisition: undersampling and non-Cartesian sampling

There is a minimum number of samples below which the measurement becomes ambiguous. It is fixed by the spacing  $\Delta k$  between two neighbouring k-space samples, which sets the extent of the region that can be reconstructed without ambiguity, the *field of view* (FOV), through  $\text{FOV} = 1/\Delta k$ . Sample too coarsely and the field of view shrinks below the size of the head; whatever falls outside it does not vanish but reappears on the opposite side of the image. This is *aliasing*, and it is why a naive acceleration produces ghosts of the anatomy superimposed on the anatomy itself.

Two families of methods make undersampling viable, and they rest on different arguments. The first, *parallel imaging*, exploits the receiving hardware. The signal is not picked up by a single antenna but by an array of coils placed around the head, and each coil is more sensitive to the region closest to it. A coil at the back of the head therefore sees a different weighting of the same object than a coil at the front. Several coils observing the same undersampled k-space provide several differently weighted views of the object, and this redundancy fills in part of what the missing samples would have told us. It comes at a price: the weighting of each coil, its *sensitivity map*, has to be known, and in practice it must be estimated from the data itself.

The second family, *compressed sensing*, rests on a property of the images themselves rather than on the hardware. A brain image looks complicated, but it contains far less information than its number of voxels suggests: large regions are homogeneous, and most of the structure lies in a small number of edges and textures. Expressed in a suitable transform domain, wavelets for instance, the image becomes *sparse*, meaning that almost all its coefficients are close to zero. If the image can be described by few coefficients, then in principle few measurements should suffice to determine it, and this is what compressed sensing formalises [3]. Recovering the image is no longer a matter of inverting the Fourier transform, but of searching among all images compatible with the measurements for the one that is sparsest.

Sparsity alone is not enough. Compressed sensing works only if the undersampling produces artefacts that look like noise rather than like structure, a property called *incoherence*. The reason is intuitive: a reconstruction algorithm that searches for the sparsest image can only succeed if the artefacts it must remove are themselves not sparse. Regular undersampling fails this test, since skipping every other k-space line produces a clean replica of the head superimposed on itself, which is just as sparse as the anatomy and cannot be told apart from it. Random or irregular undersampling, by contrast, scatters the error into a diffuse haze, which sparsity constraints can then suppress. The sampling pattern is therefore not a detail of implementation but part of the reconstruction method itself.

Where the samples are taken matters as much as how many. The energy of the k-space is not spread evenly: the centre concentrates the low frequencies, which carry the overall contrast and the large structures, while the periphery holds the high frequencies that describe edges and fine texture. The asymmetry is easy to see. Reconstructing from the central region alone gives a recognisable but blurred brain; reconstructing from the periphery alone gives edges floating in noise, with no anatomy left. The low frequencies are also the part that sparsity cannot recover, since a blurred image is not sparse in any useful sense. They must therefore be measured. This is the argument for *variable-density sampling*: acquire the centre of k-space densely, and undersample the periphery, where the missing information can be recovered at reconstruction time.

SPARKLING belongs to this family, and differs from radial or spiral trajectories in how it is built. Those are defined by a formula: their shape is fixed in advance, and only a few parameters are left to the user. SPARKLING instead treats trajectory design as an optimisation problem, first for 2D imaging [4] and later extended to full 3D [1]. The user prescribes a target sampling density, a function stating what proportion of the samples should fall in each region of k-space, and the algorithm moves the samples until the trajectory reproduces that density while spreading them apart so that no region is over-represented. Hardware feasibility is enforced throughout, since the gradients have bounded amplitude and bounded *slew rate*, the speed at which their amplitude can change. Figure 3 summarises the procedure.

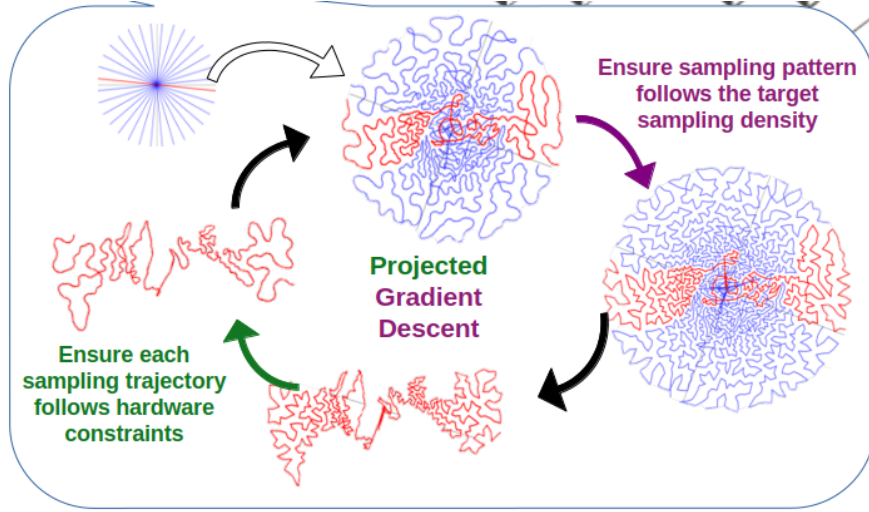


Figure 3: Principle of SPARKLING trajectory generation. A parametric trajectory, here a radial one (top left), provides the initialisation. A projected gradient descent then alternates two constraints: the sampling pattern must follow the prescribed target density, and each individual trajectory must remain feasible on the scanner. The resulting curves are smooth and irregular. Figure courtesy of Chaithya G. R.; the method is described in [4] and extended to full 3D in [1].

One property of non-Cartesian acquisitions deserves to be stated explicitly, because it becomes central later in this report. A shot is not an arbitrary grouping of samples but a physical unit: one excitation of the magnetisation, followed by one uninterrupted readout. Within a shot, consecutive samples are acquired microseconds apart along a continuous curve, so they are close in  $k$ -space, strongly correlated, and share the same excitation. Between two shots, the sequence waits for the magnetisation to recover, and the next shot starts from a fresh excitation. Any procedure that partitions the acquired data into subsets can therefore operate *within* shots, splitting the samples of each curve, or *between* shots, assigning whole curves to one subset or the other. These two options are not equivalent, and choosing between them is one of the questions this work addresses.

### 3.3 Reconstruction as an inverse problem

Reconstruction means going backwards. The scanner applies a known physical process to an unknown object and records the result; the task is to recover the object from the recording. Problems of this kind are called *inverse problems*, and they are stated by first writing down the forward process. Collecting what the previous sections established, the measurements acquired by coil  $\ell$  are

$$\mathbf{y}_\ell = \mathcal{F}_\Omega \mathbf{S}_\ell \mathbf{x} + \boldsymbol{\varepsilon}_\ell, \quad \ell = 1, \dots, L, \quad (2)$$

where  $\mathbf{x}$  is the image,  $\mathbf{S}_\ell$  the sensitivity map of coil  $\ell$ ,  $\mathcal{F}_\Omega$  the Fourier transform evaluated at the sampled locations  $\Omega$ , and  $\boldsymbol{\varepsilon}_\ell$  the acquisition noise. Reading the equation from right to left retraces the physics: the object is weighted by each coil, transformed into k-space, sampled along the trajectory, and corrupted by noise.

Written this way, reconstruction looks like solving a linear system. It is not that simple, because the system has fewer equations than unknowns. Undersampling means that the number of measurements  $M$  is smaller than the number of voxels  $N$ , so infinitely many images satisfy Equation (2) equally well. Worse, the operator amplifies noise: small perturbations in  $\mathbf{y}$  can produce large deviations in the recovered  $\mathbf{x}$ . A problem with these characteristics is said to be *ill-posed*, and no amount of numerical care will fix it, since the missing information is genuinely absent from the data. It has to be supplied from elsewhere.

The missing information is supplied as a *prior*: an assumption about what a plausible image looks like. Reconstruction is then written as a minimisation, balancing fidelity to the measurements against agreement with that assumption,

$$\hat{\mathbf{x}} = \arg \min_{\mathbf{x} \in \mathbb{C}^N} \frac{1}{2} \sum_{\ell=1}^L \|\mathbf{y}_\ell - \mathcal{F}_\Omega \mathbf{S}_\ell \mathbf{x}\|_2^2 + \mathcal{R}(\mathbf{x}), \quad (3)$$

where the first term measures how well a candidate image explains the data and  $\mathcal{R}$ , the *regulariser*, penalises images considered unlikely. Compressed sensing corresponds to a particular choice of  $\mathcal{R}$ , one that penalises the number of significant coefficients in a sparsifying transform such as a wavelet basis. Problems of this form are solved iteratively: the algorithm alternates between a step that pulls the estimate towards the measurements and a step that enforces the prior, repeating until the two are balanced.

One practical difficulty remains, specific to non-Cartesian data. Solving Equation (3) iteratively requires applying  $\mathcal{F}_\Omega$  and its adjoint many times, and the fast Fourier transform, which makes this affordable in the Cartesian case, assumes samples on a regular grid. SPARKLING samples are not on a grid. The operator is therefore implemented as a *non-uniform fast Fourier transform* (NUFFT), which interpolates between the scattered sample locations and an over-sampled grid so that a fast transform can still be used. Applying the adjoint alone gives a first, crude reconstruction, but it is biased: the centre of k-space is sampled far more densely than the

periphery, so low frequencies are counted many times over and the result is blurred. Correcting for this requires weighting each sample by the inverse of the local sampling density, an operation called *density compensation*.

### 3.4 From hand-crafted priors to learned reconstruction

The regulariser of Equation (3) encodes what we believe an image should look like, and until recently that belief was written by hand. Sparsity in a wavelet basis is a reasonable assumption, but it is a generic one: it says nothing about brains in particular, and it treats an anatomically implausible image as acceptable provided its coefficients are few. Deep learning replaces this hand-crafted assumption by one estimated from data. Given a collection of acquisitions, a network can learn what reconstructed images typically look like, and use that knowledge to fill in what undersampling removed. The gain in image quality is substantial, and learned reconstruction has become widely adopted.

The simplest way to use a network is to let it map a poor reconstruction to a better one, treating the problem as image denoising. This discards something valuable: the acquisition model is known exactly, and there is no reason to make a network rediscover it. *Unrolled networks* keep it. The idea is to take the iterative algorithm that solves Equation (3), stop it after a fixed number of iterations, and turn each iteration into a layer. The step that enforces consistency with the measurements is kept as it is, since it involves only known operators. The step that enforces the prior is replaced by a small convolutional network whose weights are learned. The result is a network whose architecture mirrors the physics of the acquisition, and which needs far fewer parameters than a generic one because it does not have to learn what is already known.

The network used in this work follows this principle. NC-PDNet [5] unrolls the primal-dual algorithm, a classical iterative scheme that alternates between the image domain and the measurement domain. Each unrolled iteration applies the NUFFT to move from one to the other, so the trajectory is built into the architecture rather than learned. Two features adapt it to non-Cartesian data: the density compensation described in Section 3.3 is applied inside the network, and the convolutional blocks operate on complex-valued images. The version used here extends NC-PDNet to three-dimensional multi-coil acquisitions [2], and its architecture is shown in Figure 4.



the network and kept aside, they can serve as a target. This is the idea behind SSDU, Self-Supervised learning via Data Undersampling [6]. The acquired k-space locations  $\Omega$  are split into two disjoint subsets,  $\Theta$  and  $\Lambda$ , with  $\Theta \cup \Lambda = \Omega$ . The network reconstructs an image from  $\Theta$  alone. The forward operator is then applied to that image to predict what would have been measured at the  $\Lambda$  locations, and those predictions are compared to the measurements actually acquired there. No reference image appears anywhere in this loop.

Formally, let  $f_\theta$  denote the reconstruction network with weights  $\theta$ , and  $\mathcal{A}_\Lambda$  the forward operator evaluated at the locations of  $\Lambda$ . Training minimises

$$\mathcal{L}(\theta) = \|\mathbf{y}_\Lambda - \mathcal{A}_\Lambda(f_\theta(\mathbf{y}_\Theta))\|_1, \quad (4)$$

where  $\mathbf{y}_\Theta$  are the measurements given to the network and  $\mathbf{y}_\Lambda$  those held out. The loss is computed in k-space, not in the image domain, since that is where the only available ground truth lives. The original formulation divides this quantity by  $\|\mathbf{y}_\Lambda\|_1$ , a normalisation that makes the loss comparable across acquisitions of different intensities. Here that role is played earlier in the pipeline, by a normalisation applied to the k-space itself, so the loss is used in the plain form above; the choice is discussed in Section 4.

Why should this work at all? The network never sees what it is meant to produce, only measurements it must be able to predict. The answer lies in what the two subsets share and what they do not. Both are drawn from the same object, so a network that reconstructs it correctly from  $\Theta$  will predict the  $\Lambda$  measurements correctly as well. But the undersampling artefacts differ: the aliasing produced by  $\Theta$  has no reason to reproduce itself at the  $\Lambda$  locations. A network that merely memorised the artefacts of its input would therefore fail on the held-out data, whereas a network that recovers the underlying object succeeds. The argument holds only as long as the two subsets carry genuinely independent information, which is why they must be disjoint. Yaman et al. show experimentally that letting  $\Theta$  and  $\Lambda$  overlap degrades reconstruction quality, and the degradation grows with the amount of overlap [6].

SSDU was formulated and validated on Cartesian acquisitions, and the split was defined accordingly. When k-space is a regular grid, the acquired locations form a mask, and partitioning them means assigning grid points to one subset or the other; Yaman et al. compare a uniform assignment with one drawn from a Gaussian distribution favouring the centre, and report better results with the latter [6]. None of this transfers directly to a non-Cartesian acquisition. There is no grid, no mask, and the samples are not free-standing points but positions along continuous curves grouped into shots. The split must therefore be defined on the trajectory itself, and as noted in Section 3.2, it can act within shots or between them. The original work explicitly leaves this extension open, and addressing it is one of the contributions of this internship.

### 3.6 Related work

Several extensions of SSDU are relevant here. Multi-mask SSDU addresses the fact that self-supervised performance degrades at high acceleration rates, where the acquired data become scarce. Drawing several different partitions from the same acquisition and averaging the resulting losses recovers part of that loss [7]. Zero-shot self-supervised learning (ZS-SSL) addresses a different question: how to train on a single scan, with no database at all [8]. It splits the acquired data three ways rather than two, the third subset serving to detect overfitting and stop training. Both remain Cartesian.

The closest work to this one is DDSS, dual-domain self-supervised learning for accelerated non-Cartesian reconstruction [9]. It supervises in both the k-space and the image domain, and it does define a split for non-Cartesian data, though how that split relates to the shot structure is not discussed. The comparison also concerns 2D single-coil or lightly parallel settings rather than the 3D multi-coil volumes considered here, where the size of the data changes what is computationally feasible.

The dataset used in this work is itself the subject of a supervised study [10], in which the training target is a compressed-sensing reconstruction of the same acquisition. Removing the need for such a target is precisely what the self-supervised formulation offers, and it is what makes the present work a direct alternative rather than a variation on that approach.

The gap this work addresses can now be stated plainly. SSDU has been validated on Cartesian acquisitions, and its splitting strategies were designed for a grid. The one existing non-Cartesian formulation splits between shots without comparing that choice to the alternative. What a suitable splitting strategy looks like for optimised non-Cartesian trajectories in 3D multi-coil imaging, and how the size and distribution of the two subsets affect the result, remain open questions. They are the ones this internship investigates.

## 4 Materials and Methods

### 4.1 The Mondor dataset

The data used in this work come from a collaboration between NeuroSpin and Henri Mondor Hospital (AP-HP), aimed at reducing the acquisition time of high-resolution susceptibility weighted imaging (SWI) in patients examined before deep brain stimulation surgery. Precise delineation of the subthalamic nucleus requires sub-millimetre resolution, which a conventional protocol cannot deliver in a clinically acceptable time. The database comprises 123 patients scanned at 3T with a 3D SWI sequence using an optimised SPARKLING trajectory. Each acquisition carries a metadata file recording which trajectory was used and to which subset of the original study the patient belongs. Four trajectories were used across the database, unevenly: 102 patients with *core*, nine with *sd1*, six each with *sd2* and *sd3*. The patients are partitioned into 99 for training, 11 for validation, 11 for testing, and two excluded. The work reported here uses the *sd1* subset, which contains seven training patients, one validation patient, and one test patient, number 006. Restricting the study to a single trajectory avoids having to handle several sampling geometries at once; using the original partition means the held-out patient is the one the reference study also held out, rather than an arbitrary choice.

Each volume covers a field of view of  $384 \times 384 \times 208$  voxels at 0.6 mm isotropic resolution, acquired with five receiver coils. The trajectory consists of 8192 shots of 2048 samples each, giving  $1.68 \times 10^7$  k-space samples per volume, with coordinates normalised to  $[-0.5, 0.5]^3$ . The echo time is 20 ms and the readout duration  $T_{obs} = 20.48$  ms. Three files are available for each patient: the measured multi-coil k-space, a set of refined coil sensitivity maps, and an estimated  $\Delta B_0$  field map expressed in Hz.

No fully sampled acquisition exists for these patients: the k-space was undersampled by a factor of about 20, and a complete acquisition was never performed. There is therefore no ground truth against which reconstructions could be scored. What the dataset provides is a compressed-sensing reconstruction of each volume, obtained with a sparse-threshold algorithm run for 20 iterations [10], in two variants: NOC, computed without off-resonance correction, and COR, computed with it. These serve as the point of comparison in this report, with the important qualification that they are reconstructions rather than references. Since the pipeline described below performs no off-resonance correction, comparisons are made against NOC.

One further difference must be stated. The available reconstructions were computed from the full twenty-channel acquisition, whereas the pipeline described here operates on a five-channel compression of the same data. No five-channel reconstruction exists, so part of any discrepancy observed later originates in the channel compression rather than in the reconstruction method itself.

## 4.2 k-space normalisation

The raw k-space magnitudes are small, small enough that gradients computed in single precision lose numerical resolution during training. An early version of the pipeline showed this directly: the loss failed to decrease, and rescaling the data resolved it. The k-space is therefore normalised before anything else happens.

The scaling factor is derived from the centre of k-space rather than from a global statistic. Writing  $\mathcal{C}$  for the set of samples closest to the centre, the factor is the mean multi-coil magnitude over that region,

$$s = \frac{1}{|\mathcal{C}|} \sum_{k \in \mathcal{C}} \sqrt{\sum_{\ell=1}^L |y_{\ell}(k)|^2}, \quad \tilde{\mathbf{y}} = \mathbf{y}/s. \quad (5)$$

Two reasons motivate this choice over a global percentile. The centre of k-space carries the low-frequency, high-energy part of the signal, whose physical meaning does not depend on the trajectory, which makes the factor comparable across acquisitions. And because that region is densely sampled in every SPARKLING trajectory, the estimate is stable from one patient to the next. This normalisation is also what allows the loss of Equation (4) to be used without the per-acquisition division of the original formulation.

## 4.3 Splitting the acquisition into $\Theta$ and $\Lambda$

Section 3.5 left the question open: on a non-Cartesian trajectory, what exactly should be split? The acquisition has a structure that a Cartesian mask does not, since its samples are grouped into shots, and two partitions are therefore possible. An *inter-shot* split assigns whole shots to one subset or the other, so a shot is either seen by the network or held out for the loss. An *intra-shot* split divides the samples of every shot between the two subsets, so each shot contributes to both. The strategy implemented here is the second.

Concretely, for each of the 8192 shots, the 2048 sample positions are randomly permuted, and a fixed proportion is assigned to  $\Theta$  and the remainder to  $\Lambda$ . With a 60/40 ratio this yields 1229 samples per shot for  $\Theta$  and 819 for  $\Lambda$ , that is  $1.01 \times 10^7$  and  $6.72 \times 10^6$  samples respectively. The two subsets are disjoint by construction, as SSDU requires, and each of them covers the whole extent of k-space: because every shot contributes to both,  $\Theta$  and  $\Lambda$  approximately inherit the variable density of the original trajectory rather than concentrating in different regions of k-space.

Whatever the geometry, the partition must satisfy four conditions. The two subsets must cover the acquisition and not overlap,  $\Theta \cup \Lambda = \Omega$  and  $\Theta \cap \Lambda = \emptyset$ ; the prescribed fraction must be respected; and the sampling density of each subset must match that of the full acquisition,  $\rho_{\Theta}(k) = \rho_{\Lambda}(k) = \rho_{\Omega}(k)$ . The first three are enforced by any correct partition. The fourth is the one that distinguishes strategies, and it matters here for a reason specific to non-Cartesian

imaging: the trajectory was optimised to realise a prescribed density, and it is that density which makes the encoding operator well conditioned. A partition that disturbs it hands the network an input whose sampling profile no longer matches what the acquisition was designed for, degrading the reconstruction for a reason unrelated to self-supervision.

The per-shot selection described above satisfies the fourth condition in expectation but not exactly: every shot yields the same number of samples, yet nothing constrains how those samples distribute across k-space radii, and each draw fluctuates around the target density. An alternative is to stratify by radius instead of by shot, grouping the samples into radial bins and drawing the same proportion within each bin. The density is then preserved by construction rather than on average. The two schemes guarantee different things and leave the other free: the first fixes the per-shot proportion and lets the radial density fluctuate, the second fixes the density and lets the per-shot proportion vary. Section 5.4 quantifies the difference.

Figure 5 shows what the network is given. Applying the adjoint operator to  $\Theta$  alone, with the coil sensitivity maps, produces a complex volume in which the anatomy is already legible despite using 60% of the acquired samples, with visible noise and residual artefacts. This volume, not the k-space itself, is what enters the first unrolled iteration.

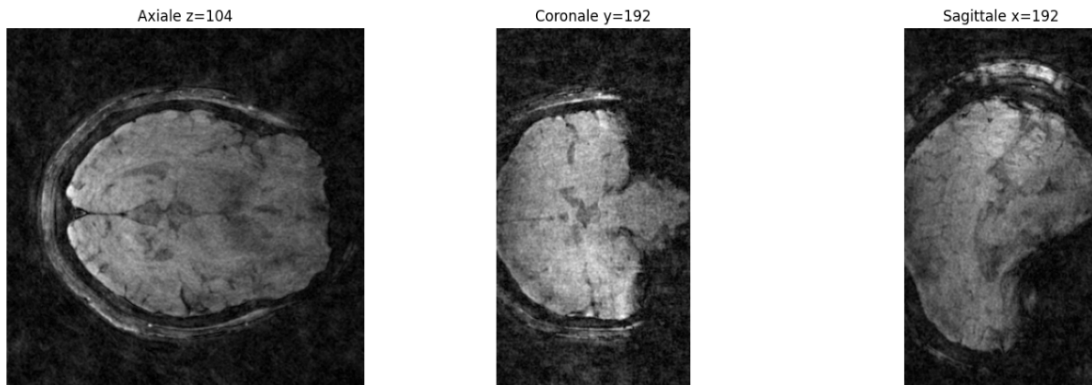


Figure 5: Adjoint reconstruction from the  $\Theta$  subset alone, that is from 60% of the acquired samples, shown for patient 001 in the three orthogonal planes. Coil sensitivity maps are applied during the adjoint operation. The anatomy is preserved, with noise and residual undersampling artefacts. This complex-valued volume is the input to NC-PDNet.

A note on notation. The split is parameterised here by the fraction of samples given to the network,  $|\Theta|/|\Omega|$ , which is the quantity the implementation takes as input. Yaman et al. parameterise it by the complementary fraction,  $\rho = |\Lambda|/|\Omega|$ , and report  $\rho = 0.4$  as their default; the 60/40 ratio used here is the same choice expressed the other way round.

#### 4.4 Network configuration and training

The implementation of NC-PDNet used here is the three-dimensional multi-coil version developed by Asma Tanabene [2]. The network is unrolled over six iterations, with a buffer size of two and 32 convolutional filters per block, and produces a complex-valued image. This amounts

to 217,728 trainable parameters, a small figure by deep-learning standards, which reflects the fact that the acquisition model is built into the architecture rather than learned. The non-uniform Fourier operators are provided by the `mri-nuffft` library with the `gpunuffft` backend. The density compensation weights used by the operators restricted to  $\Theta$  and  $\Lambda$  accompany the trajectory and were pre-computed by G. Daval-Fr  rot; those internal to NC-PDNet are recomputed by the network from the sampling locations it is constructed with.

At each training step, the network receives the  $\Theta$  measurements together with the coil sensitivity maps, and returns a complex volume. The forward operator restricted to  $\Lambda$  is applied to that volume, and the loss of Equation (4) compares the result with the held-out measurements. Gradients propagate back through the NUFFT operators, which are differentiable, so the whole pipeline is trained end to end.

Two training regimes are compared. In the first, the partition is drawn once before training and kept fixed, so the network sees the same  $\Theta$  at every epoch. In the second, a new partition is drawn at the start of each epoch. This has a cost specific to the non-Cartesian setting: the NUFFT operator is built from the sampling locations, so changing  $\Theta$  requires rebuilding both the operator restricted to  $\Lambda$  and the network itself, the trajectory being fixed at construction. Network weights and optimiser state are carried across, the architecture being unchanged. Rebuilding costs about 19 s per epoch, which raises a hundred-epoch run from roughly 26 to 65 minutes. On a Cartesian grid the same operation would be free, since changing which lines are kept does not change the size of the FFT.

## 4.5 Evaluation protocol

The split into  $\Theta$  and  $\Lambda$  serves training only. At evaluation time the network must be given every sample that was acquired, since withholding data at test time would penalise the method for no reason. Yaman et al. state this explicitly [6], and it has a practical consequence in the present implementation. NC-PDNet fixes its internal NUFFT operator when the model object is constructed: the trajectory is passed to the constructor and stored, and the forward pass does not accept a different one. A model trained on  $\Theta$  therefore cannot simply be handed the full k-space. A second model is instead built with the complete trajectory  $\Omega$ , and the weights learned during training are transferred into it before inference.

Reconstruction quality is quantified against the NOC reference using the peak signal-to-noise ratio (PSNR) and the structural similarity index (SSIM). Both are computed on magnitude images. It should be kept in mind that these figures measure agreement with another reconstruction rather than with a true image, so they indicate how closely the two methods agree, not how accurate either of them is in absolute terms.

Two evaluations are distinguished throughout. Measuring on the volume the network was trained on says whether the optimisation succeeded, not whether the reconstruction generalises. Two volumes are therefore held out entirely: patient 006, which belongs to the test set of the

original study, and patient 002, which belongs to its validation set. Both share the trajectory of the training volumes and neither is seen during training. Configurations are compared on those same volumes, so that differences can be attributed to the training strategy rather than to variation in test data.

## 4.6 Implementation and computing environment

The pipeline is written in Python and packaged as an installable library, `ssdu_mondor`, organised into modules that separate data handling, model construction and training. Configuration is managed with Hydra, so that every run records the exact parameters it was launched with; this matters for a study whose purpose is to compare configurations, since a result is only interpretable if the settings that produced it are known. The source code is versioned in a private repository.

Training runs on the Topaze cluster of the CCRT, on NVIDIA A100 GPUs, with jobs submitted through SLURM. The scale of the data is the main practical constraint: a single volume holds  $1.68 \times 10^7$  k-space samples across five coils, and each unrolled iteration applies a non-uniform Fourier transform to a  $384 \times 384 \times 208$  volume. Memory, rather than computation time, is what limits the configurations that can be trained.

## 4.7 Off-resonance correction

The forward model of Equation (2) assumes that the static magnetic field is perfectly uniform across the head. It is not. Differences in magnetic susceptibility between tissue, bone and air shift the resonance frequency locally, by an amount described by a field map  $\Delta B_0$  expressed in Hz. During a readout, this shift accumulates an extra phase that grows with time, so the signal no longer matches the pure Fourier model assumed by the reconstruction. In Cartesian imaging the effect is mostly a geometric distortion, but non-Cartesian readouts are long and sweep k-space in all directions, which turns the same effect into blurring. Susceptibility weighted imaging is particularly exposed, since it relies on long echo times and on the phase of the signal.

Correcting for this means extending the forward operator so that it incorporates the field map, an approach developed within the team by G. Daval-Fr  rot [11, 10], and the Mondor dataset includes an estimated field map for every patient. Integrating it into the present pipeline was attempted but is not part of the results reported here. The extended operator approximates the phase term by a low-rank decomposition, and the interpolator built for that purpose proved too memory-intensive to coexist with an unrolled network on a single GPU: training stopped with an out-of-memory error at the first step. Reducing the resolution of the interpolator brought the memory footprint down without resolving the problem. Off-resonance correction is therefore left aside in what follows, and all results are obtained without it, which is also why the NOC variant of the reference reconstruction is the one used for comparison.

## 5 Results

### 5.1 Training behaviour

The first question is whether the self-supervised objective trains at all in this setting. Figure 6 shows the loss over 100 epochs for a network trained on a single volume, patient 001, without off-resonance correction. The loss decreases by about 5% over the first sixty epochs and then rises slightly, which indicates that the optimisation is well behaved: gradients propagate through the NUFFT operators, and the network does learn to predict the held-out  $\Lambda$  measurements from the  $\Theta$  ones.

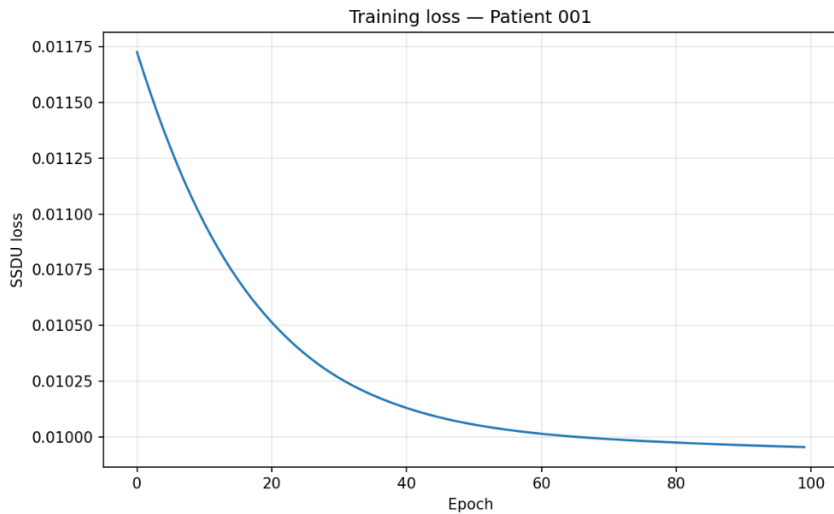


Figure 6: SSDU training loss over 100 epochs, for a network trained on a single volume without off-resonance correction. The loss decreases by about 5% over the first sixty epochs and then rises slightly.

This curve says nothing about generalisation. Training on a single volume, with a partition drawn once and kept fixed throughout, the network is free to specialise on that particular acquisition and on that particular split. A decreasing loss under these conditions is a necessary condition for the pipeline to be correct, not evidence that the reconstruction is good.

### 5.2 Comparison with the reference reconstruction

Figure 7 places the reconstruction obtained with SSDU next to the NOC reconstruction of the same patient. The anatomy is recovered: the ventricles, the cortical folding and the skull are in the right places, and the overall contrast is comparable. The self-supervised network has therefore learned something about the object rather than merely reproducing the aliasing of its input, which is the property the argument of Section 3.5 predicts. The reconstruction is nonetheless granular, the grey-white matter contrast is weaker than in the reference, and the residual error concentrates along the skull and the cerebral periphery, which points to a difference in sharpness rather than a structural error.

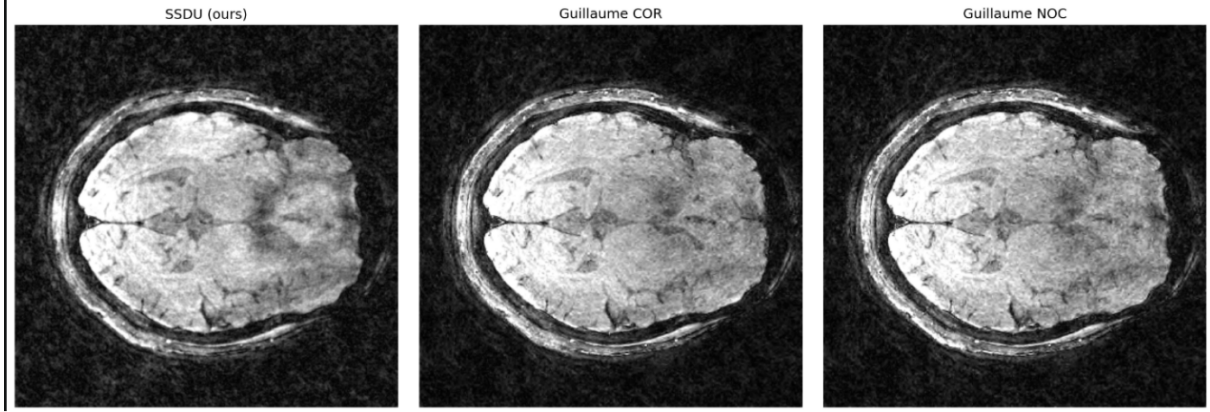


Figure 7: Axial slice of patient 001. Left: reconstruction obtained with the SSDU pipeline, without off-resonance correction. Centre and right: the compressed-sensing reconstructions of the same acquisition, with (COR) and without (NOC) off-resonance correction. NOC is the variant used for comparison throughout, since the present pipeline performs no off-resonance correction. All three recover the same anatomy; the SSDU reconstruction is granular and its grey-white matter contrast is weaker.

Quantifying the agreement requires deciding where to measure it. Computed over the whole volume, both metrics are dominated by the air surrounding the head, which contains no signal and on which the two reconstructions trivially agree; the PSNR is inflated as a result. A brain mask was therefore derived from the reference volume, using a threshold at a quarter of its 99th percentile, and retains 30.4% of the voxels. The same mask is used for every run on a given patient, so that the figures remain comparable. Table 1 gives both readings.

Table 1: Agreement between the SSDU reconstruction and the NOC reconstruction, for patient 001, trained for 100 epochs at a 60/40 split ratio.

| Region            | PSNR (dB) | SSIM  |
|-------------------|-----------|-------|
| Whole volume      | 23.05     | 0.620 |
| Within brain mask | 19.98     | 0.806 |

The masked figures are the meaningful ones, and they are used throughout the remainder of this report. Their absolute level should be read with the two qualifications set out in Section 4: the comparison is against another reconstruction rather than a ground truth, and it is made across a difference in channel count that no available data allow to isolate.

### 5.3 Effect of the split ratio

The proportion of samples assigned to  $\Theta$  is a free parameter of the method, and the value used so far, 60/40, was inherited rather than justified. To test its influence, the same network was trained five times on the same volume, with  $|\Theta|/|\Omega| \in \{0.4, 0.5, 0.6, 0.7, 0.8\}$  and everything else held fixed. Two of these runs did not complete: at  $|\Theta|/|\Omega| = 0.7$  and  $0.8$  the job stopped

with an out-of-memory error inside the non-uniform Fourier operator. This is a consequence specific to the non-Cartesian setting, where the size of the operator depends on the number of sampled locations; on a Cartesian grid, changing which lines are kept does not change the size of the FFT.

Table 2: Reconstruction quality as a function of the fraction of samples assigned to  $\Theta$ , measured within the brain mask against the NOC reconstruction, patient 001, 100 epochs. The last row repeats the 0.6 configuration to estimate run-to-run variability.

| $ \Theta / \Omega $ | PSNR (dB) | SSIM   |
|---------------------|-----------|--------|
| 0.4                 | 19.99     | 0.8054 |
| 0.5                 | 20.17     | 0.8079 |
| 0.6                 | 20.10     | 0.8079 |
| 0.6 (repeat)        | 19.98     | 0.8056 |

The three completed runs differ by 0.0025 in SSIM. Repeating the 0.6 configuration changes the result by 0.002, so the spread between ratios is of the same order as the variability between two runs of the same configuration. No ratio can be preferred on this evidence.

This is a negative result, and it is worth stating why it was expected. The loss decreased by only about 5% before rising again from epoch 60 onwards, the signature of a network memorising a single acquisition and a single fixed partition of it. Under those conditions the reconstruction is limited by overfitting, not by how the samples are divided, and the ratio has little room to make a difference. Establishing the run-to-run variability was nevertheless necessary: it sets the threshold, about 0.002 in SSIM, below which no difference reported in this work should be considered meaningful.

## 5.4 Preserving the sampling density within the split

The ratio is not the only degree of freedom. How the samples are chosen within each shot also matters, and the argument of Section 3.2 suggests a requirement: since the trajectory was optimised to follow a prescribed density, a split that disturbs that density gives the network an input whose sampling profile no longer matches what the acquisition was designed for.

The density of the trajectory used here was measured directly. It follows a power law in  $\|k\|^{-0.84}$  rather than the Gaussian shape that the selection probability of the Cartesian formulation of SSDU implicitly favours; a Gaussian fit to the measured profile is degenerate. A selection procedure that does not presuppose any particular shape is therefore preferable. Samples are grouped into radial bins, and within each bin the same proportion is drawn at random, so that the density of  $\Theta$  reproduces that of  $\Omega$  by construction whatever its shape.

Measured as the deviation between the density of  $\Theta$  and that of the full trajectory, the improvement is substantial:  $3.31 \times 10^{-4}$  for the per-shot selection used until now, against  $1.11 \times 10^{-6}$  for radial stratification, a factor of about 300. The procedure remains intra-shot,

as intended: no shot ends up entirely in one subset, the per-shot fraction assigned to  $\Theta$  staying between 56% and 64%.

One incidental finding is worth recording. The nominal and effective ratios differed slightly, a nominal  $|\Theta|/|\Omega| = 0.6$  yielding 0.5996 in practice, because the number of samples per shot was truncated rather than rounded. The discrepancy is too small to affect any result reported here, but it was corrected. Whether this better-controlled partition produces a better reconstruction is a separate question.

## 5.5 Effect of the splitting geometry

The stratification of Section 5.4 improves the density of  $\Theta$  by a factor of three hundred, but that is a property of the partition, not of the image it leads to. To find out whether it changes anything, the same network was trained twice on patient 001, for 100 epochs at the same ratio, differing only in how the samples were selected within each shot.

Table 3: Reconstruction quality for the two intra-shot selection schemes, measured within the brain mask against the NOC reconstruction, patient 001, 100 epochs,  $|\Theta|/|\Omega| = 0.6$ .

| Selection within each shot | PSNR (dB) | SSIM   |
|----------------------------|-----------|--------|
| Uniform per shot           | 19.98     | 0.8060 |
| Radially stratified        | 20.00     | 0.8062 |

The two reconstructions differ by 0.0002 in SSIM, an order of magnitude below the variability of 0.002 established in Section 5.3. The geometry of the split, like its ratio, cannot be evaluated on this evidence.

The training curves do differ, and in a way worth recording. The loss of the stratified run decreases by 8.2% against 5.4% for the uniform one, and it decreases monotonically over the full hundred epochs where the uniform run begins rising again from epoch 60. The two losses are computed on different  $\Lambda$  subsets, of different sizes, so their absolute values are not comparable: the self-supervised objective is bounded below by a noise floor proportional to  $|\Lambda|$ . Only the shape of the curves is informative. Read that way, the absence of the late rise suggests that preserving the sampling density gives the network a more regular objective. It remains an indication, since no measurable difference appears in the reconstruction.

What can be stated is that the stratified selection costs nothing. It delivers a partition whose density is controlled by construction rather than in expectation, at no expense in image quality, which is reason enough to prefer it as a default.

## 5.6 Generalisation to unseen patients

Every figure reported so far was measured on the volume the network was trained on. That is a measure of fit, not of reconstruction quality, and the discussion of Section 5.3 turned on

the suspicion that the network had memorised its single training acquisition. The metadata described in Section 4.1 make it possible to test this directly: patient 006 belongs to the test set of the original study and patient 002 to its validation set. Both share the trajectory of patient 001, and neither was seen during training which is what the measurement requires.

The weights trained on patient 001 were transferred, unchanged, into a model built on the full trajectory and applied to each of the two held-out acquisitions. The run used here draws a new partition at each epoch, a variant introduced in Section 5.7; the fixed-partition run gives the same picture, as Table 5 shows.

Table 4: The same network, trained on patient 001 for 100 epochs with radially stratified splitting and a partition redrawn at every epoch, evaluated on the volume it was trained on and on two volumes held out from training. All figures are measured within a brain mask derived from the corresponding NOC reconstruction.

| Evaluated on | Seen in training | PSNR (dB) | SSIM   |
|--------------|------------------|-----------|--------|
| Patient 001  | yes              | 19.95     | 0.8043 |
| Patient 002  | no               | 18.25     | 0.7701 |
| Patient 006  | no               | 18.02     | 0.7656 |

The two held-out volumes score 0.7701 and 0.7656, within 0.005 of one another, while the training volume scores 0.8043. The reconstruction therefore loses about 0.038 in SSIM and 1.8 dB in PSNR on data it has not seen. Unlike every comparison reported earlier in this section, this difference is unambiguous: it is nearly twenty times the run-to-run variability established in Section 5.3.

That the two held-out volumes agree to within 0.005 matters. Patient 006 carries a large hypo-intense lesion absent from the training volume, visible in the reference reconstruction as well as in ours, and structural metrics are sensitive to such a difference in content; patient 002 carries no comparable lesion. Had the gap been an artefact of that lesion, the two volumes would not have scored alike. They do, which establishes that the gap is a property of generalisation rather than of one particular acquisition.

Two further checks rule out trivial explanations. The normalisation factors of the three acquisitions agree to within 1.4%, so the network is not being fed data at an unfamiliar scale. And the brain masks cover 30.4%, 30.2% and 30.3% of their respective volumes, so the three are not scored over regions of different extent.

This is the measurement the earlier negative results were missing. The ablations on the split ratio and on the splitting geometry both returned differences below the noise floor, and the explanation offered in each case was that a network trained on a single fixed acquisition is limited by overfitting rather than by how its samples are divided. Table 4 quantifies that overfitting. It also sets the target for what follows: any change to the training protocol should be judged on the held-out volumes, where there is a gap of 0.038 in SSIM to recover, and not on the training volume, where there is almost nothing left to gain.

## 5.7 Reshuffling the partition between epochs

Section 5.6 established that the network overfits its single training volume. Two remedies were available: drawing a new  $\Theta/\Lambda$  partition at every epoch, so that the network never sees the same subset twice, and training across several volumes. They were tested separately, starting with the first.

Reshuffling has a cost in this setting, described in Section 4: the NUFFT operator is built from the sampling locations, so a new partition requires rebuilding both the operator restricted to  $\Lambda$  and the network itself. Measured at 19 s per epoch, this raises a hundred-epoch run from 26 to 65 minutes.

Table 5: Reconstruction quality with a partition drawn once and kept fixed, against a partition redrawn at every epoch. Patient 001, 100 epochs, radially stratified splitting,  $|\Theta|/|\Omega| = 0.6$ . Both configurations are evaluated on the training volume and on the held-out volume.

| Partition  | Patient 001 | Patient 006 | Gap    |
|------------|-------------|-------------|--------|
| Fixed      | 0.8062      | 0.7673      | 0.0389 |
| Reshuffled | 0.8043      | 0.7656      | 0.0387 |

Neither figure improves, and the generalisation gap is unchanged: 0.0387 against 0.0389, a difference of  $2 \times 10^{-4}$ . The small drop in both absolute scores, about 0.0018, sits at the edge of the variability established in Section 5.3 and cannot be called a degradation either.

The training curves agree. Both runs converge to the same value, 0.009939 with reshuffling against 0.009940 without, which is the noise floor  $\sigma^2|\Lambda|$  that Section 3.5 identifies as the lower bound of the self-supervised objective. Two networks reaching the same floor by different routes produce the same images. The reshuffled run reaches its minimum at epoch 59 rather than epoch 99, and its loss is marginally less variable across epochs,  $1.6 \times 10^{-4}$  against  $2.3 \times 10^{-4}$  in standard deviation. The latter is worth noting in passing: a partition redrawn at every epoch might have been expected to make the loss noisier, and it does not, which indicates that the stratified draws of Section 5.4 are statistically equivalent to one another.

The result is negative and it is informative. Reshuffling changes which samples the network sees; it does not change which anatomy it sees. The overfitting measured in Section 5.6 is to a single volume, and resampling within that volume cannot address it. The remaining remedy, training across several patients, acts on the variable that actually varies.

## 6 Discussion

### 6.1 What the results establish, and what they do not

The pipeline works in the sense that mattered first: a self-supervised objective can be optimised end to end on 3D multi-coil non-Cartesian data, gradients propagate through the non-uniform Fourier operators, and the resulting network produces anatomically plausible volumes from data it was never shown a reference for. Given that no fully sampled acquisition exists for this dataset, and that the supervised study built on it had to use a compressed-sensing reconstruction as its target [10], this is the property that makes the approach worth pursuing.

What the results do not establish is that the reconstruction is good. Every figure reported here comes from a network trained on one volume, and the loss curve shows the expected consequence: a decrease of less than ten per cent, reaching a floor that the decomposition of Section 3.5 identifies as the noise term  $\sigma^2|\Lambda|$ . The network is fitting an acquisition rather than learning a prior over brains.

The extent of that fitting is now measured. The same weights score 0.8043 in SSIM on the training volume and 0.7701 and 0.7656 on two volumes never seen during training, a gap of about 0.038 against a run-to-run variability of 0.002. This is the only unambiguous difference reported in this work, and it explains why the others were not: the ablation on the split ratio and the comparison between splitting geometries both returned differences of the order of 0.002, because a parameter can only be seen to matter once it is no longer the limiting factor.

### 6.2 Practical constraints of the non-Cartesian setting

Three obstacles encountered during this internship are specific to non-Cartesian data, and all three concern memory rather than computation time.

The first appeared in the ratio study: at  $|\Theta|/|\Omega| = 0.7$  and above, training stopped with an out-of-memory error inside the non-uniform Fourier operator. Unlike a Cartesian mask, which selects entries of a fixed-size transform, the non-uniform operator is built around the sampled locations themselves, so its footprint grows with the size of  $\Theta$ . The ceiling lies between 10.1 and 13.4 million samples on an A100, and it is reached well before the physics of the problem becomes the obstacle.

The second is the same constraint seen from another angle, and it limits how much data can be used at all. Of the 123 volumes in the database, 102 were acquired with a trajectory holding 50.2 million samples, three times the 16.8 million of the one used here. At a 60/40 ratio the encoding operator for  $\Theta$  would be built on 30 million locations, more than twice the point at which training already fails. The study is therefore confined to the nine volumes sharing the present trajectory: seven for training, one for validation and one for test.

The third concerns off-resonance correction, described in Section 4. Extending the forward operator with the field map requires an interpolator whose memory footprint could not be

reconciled with an unrolled network on a single device, and reducing its resolution did not resolve the problem. This is why all results reported here are without off-resonance correction, and why the comparison is made against the NOC variant of the reference reconstruction.

Taken together, these three make memory the binding constraint of the project. It is worth stating plainly, because it is not what one expects when moving a self-supervised method from Cartesian to non-Cartesian data: the difficulty is not that the loss behaves differently, but that the operator does.

### **6.3 Limits of the comparison**

The figures reported in Section 5 measure agreement between two reconstructions, and this framing deserves to be taken seriously rather than treated as a caveat. The NOC volume is a compressed-sensing reconstruction of the same undersampled acquisition, produced by an algorithm with its own assumptions and its own failure modes. A reconstruction that departed from it would not thereby be wrong, and one that matched it perfectly would only have reproduced it, including whatever it gets wrong. In the absence of a fully sampled acquisition, no metric computed on this dataset can distinguish accuracy from agreement.

A second limitation is more concrete. The available reconstructions were computed from twenty channels, whereas the pipeline described here operates on a five-channel compression of the same data, and no five-channel reconstruction exists. Part of the discrepancy therefore originates in the channel count rather than in the reconstruction method, and the present data do not allow the two contributions to be separated. Producing a five-channel reference would isolate them, and would make every figure in this report interpretable in a way they currently are not.

A third point concerns the held-out volumes. Only one patient in the test set of the original study shares the trajectory used here, and that volume carries a large hypo-intense lesion absent from the training data, to which structural metrics are sensitive. Adding the validation volume, which carries no comparable lesion, resolves the ambiguity: the two score within 0.005 of each other, so the measured gap reflects generalisation rather than the difficulty of one acquisition. With only two held-out volumes available on this trajectory, however, the estimate remains coarse, and no confidence interval can be attached to it.

### **6.4 Directions for the remainder of the work**

Three questions follow from what precedes, and they are ordered by what currently blocks them.

Training on several patients is the first, because it removes the factor that limits everything else. Once the network sees more than one acquisition, it can no longer succeed by fitting a single one, and the parameters studied here regain the room to make a measurable difference. Seven training volumes are available on the present trajectory, which is few but is not one. Extending beyond them means either reducing the memory footprint of the encoding operator,

through gradient checkpointing or a coarser interpolation, or accepting a smaller  $\Theta$  than the ratio study suggests.

Redrawing the partition between epochs was the second, and it has now been tested. It costs 19s per epoch in this setting, since the Fourier operator must be rebuilt from the new sampling locations, which raises a hundred-epoch run from 26 to 65 minutes. It changes nothing measurable: neither reconstruction quality nor the generalisation gap moves, and both runs converge to the same noise floor. The reason is visible once stated: reshuffling changes which samples the network sees, not which anatomy, and the fitting measured here is to a single volume. Multi-mask SSDU reports a benefit from averaging over partitions [7], but in a setting where the network already sees many acquisitions; on a single volume the two variations are not equivalent.

The comparison between intra-shot and inter-shot splitting is the third, and it is the question this work set out to address. The intra-shot variant was implemented in two forms, and the density-preserving one improves the sampling profile of  $\Theta$  by two orders of magnitude at no cost in image quality, which is reason enough to adopt it. Whether inter-shot splitting does better remains open, and it carries a genuine trade-off: it restores the independence between the noise in  $\Theta$  and in  $\Lambda$  that the self-supervised objective assumes, since distinct shots are acquired in distinct repetition intervals, but it removes whole shots and so degrades the angular coverage the trajectory was optimised to provide. Deciding between them requires a network that is no longer limited by a single training volume, since a difference of the order of 0.002 in SSIM cannot be resolved against a gap of 0.038.

## 6.5 Personal assessment

This internship was my first contact with research as it is actually practised, and what surprised me most is how much of it consists of establishing that a result means what one thinks it means. The ablation on the split ratio is the clearest example: the figures were produced quickly, and the work that followed, repeating a configuration to measure run-to-run variability, was what made them interpretable. Without it I would have reported a best ratio that was indistinguishable from noise.

The same reflex settled the generalisation result. A single held-out volume would have left the gap open to a simpler explanation, since that patient carries a lesion the training volume does not; evaluating a second one, free of any comparable lesion, is what turned a suggestive number into a conclusion.

Several other habits came out of that. Verifying a claim in the source code rather than assuming it, which is how the constraint on the fixed NUFFT operator was found and the evaluation protocol corrected. Reporting a negative result as a result, which is what both the ratio ablation and the reshuffling experiment turned out to be. Stating explicitly what a metric does and does not measure, which is why the comparison against NOC is framed as agreement

rather than accuracy.

On the technical side I learned to work on a computing cluster, to structure a project as an installable package with reproducible configuration, and to diagnose failures that produce no error message, of which there were several: weights written to the wrong directory by a silent change in default behaviour, a partition ratio quietly truncated rather than rounded, and a memory leak that only appeared at the third epoch.

I also learned to check what is actually available before assuming a limitation. The metadata I believed missing, and which I had described as blocking multi-patient training, were on the shared filesystem all along; the constraint that genuinely blocks it, the size of the alternative trajectory, only became visible once I looked at the file rather than reasoning about it.

What the team gains from this work is a self-supervised pipeline for 3D multi-coil non-Cartesian reconstruction that runs end to end, a splitting procedure that preserves the sampling density of an optimised trajectory, a protocol for measuring generalisation on held-out volumes, and a documented account of the constraints that limit all of it, memory foremost among them. The questions left open are formulated precisely enough to be taken up directly.

## 7 Conclusion and Perspectives

Deep learning has become the dominant approach to accelerated MRI reconstruction, but its usual form requires fully sampled reference images, and those are precisely what high-resolution 3D imaging cannot provide. The dataset studied here illustrates the difficulty: acquired at 0.6 mm isotropic resolution and undersampled by a factor of about 20, it has no ground truth, and the supervised study built on it had to train against a compressed-sensing reconstruction of the same data.

This internship implemented SSDU for that setting. The acquired k-space is split into two disjoint subsets, one given to the network and one held out to compute the loss, so that training requires no reference image at all. Transposing the method from Cartesian acquisitions, for which it was designed, to optimised non-Cartesian trajectories raised a question the original work leaves open: what exactly should be split, when the samples lie along continuous shots rather than on a grid.

The strategy implemented divides the samples within each shot. A stratified variant was then developed that reproduces the sampling density of the full trajectory to within  $10^{-6}$ , against  $3 \times 10^{-4}$  for the scheme it replaces, and does so without assuming any functional form for that density which matters here, since the measured profile follows a power law rather than the Gaussian shape the Cartesian formulation implicitly favours through its selection probability. The improvement is in the partition and not in the images: reconstruction quality is unchanged, and the case for the scheme is that it costs nothing while replacing a quantity left to chance by one fixed by construction.

The pipeline trains end to end and produces anatomically plausible volumes from data it was never shown a reference for. It does not yet reconstruct well, and the extent of the shortfall is now measured. A network trained on a single acquisition scores 0.8043 in SSIM on that volume and 0.7701 and 0.7656 on two volumes it has never seen, a gap of 0.038 against a run-to-run variability of 0.002. Establishing that variability was itself necessary: it sets the threshold below which no comparison in this work should be believed, and it explains why the ablations on the split ratio and on the splitting geometry both returned differences too small to interpret. A parameter can only be seen to matter once it is no longer the limiting factor.

Redrawing the partition at every epoch was the obvious remedy and it does not work. It costs 19 s per epoch, since the Fourier operator must be rebuilt from the new sampling locations, and it moves neither the reconstruction quality nor the generalisation gap. The reason is visible once stated: reshuffling changes which samples the network sees, not which anatomy, and what the network has fitted is the anatomy of one volume.

What remains is therefore ordered by that finding. Training across several patients is the lever that acts on the right variable, and seven volumes sharing the present trajectory are available for it. Extending beyond them is constrained by memory rather than by data: the trajectory used for the majority of the database holds three times more samples, which would place the

encoding operator well past the point at which training already fails on an A100. Only once the network is no longer limited by a single volume does the comparison between intra-shot and inter-shot splitting become measurable, and with it the question this work set out to answer. The groundwork for that comparison, the splitting schemes, the evaluation protocol, and the threshold against which any difference must be read is in place.

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